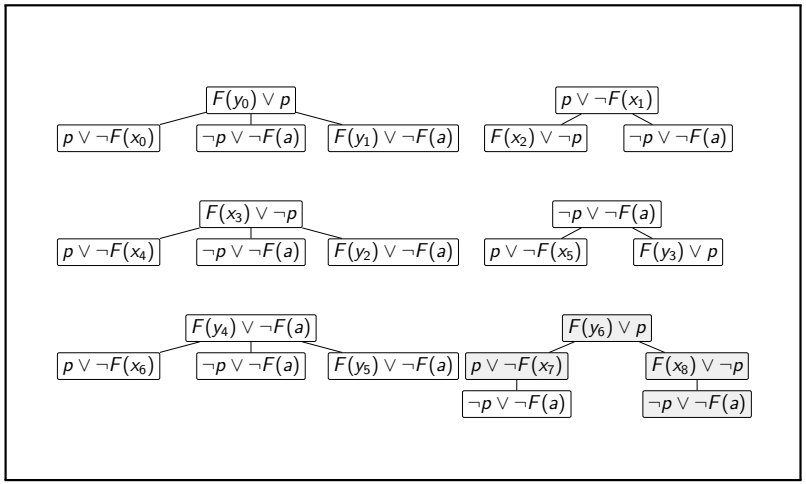
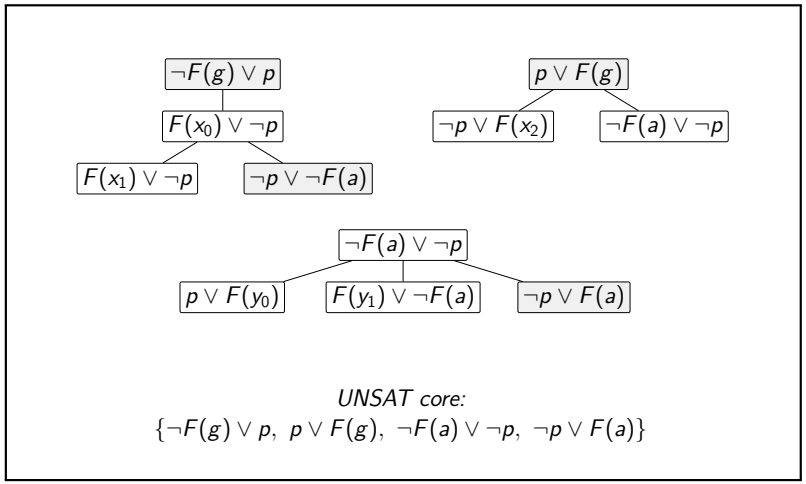
Previous Work

SATCoP (Rawson and Reger 2021) improves over LeanCoP by using a SAT solver as unsat proof.

SYN052-1.p

$$p \vee \neg F(X) \qquad F(X) \vee \neg p$$
$$F(Y) \vee p \qquad \neg p \vee \neg F(a) \qquad F(Y) \vee \neg F(a)$$





LeanCoP: unsat proof in **43 steps** by closing the Tableau.

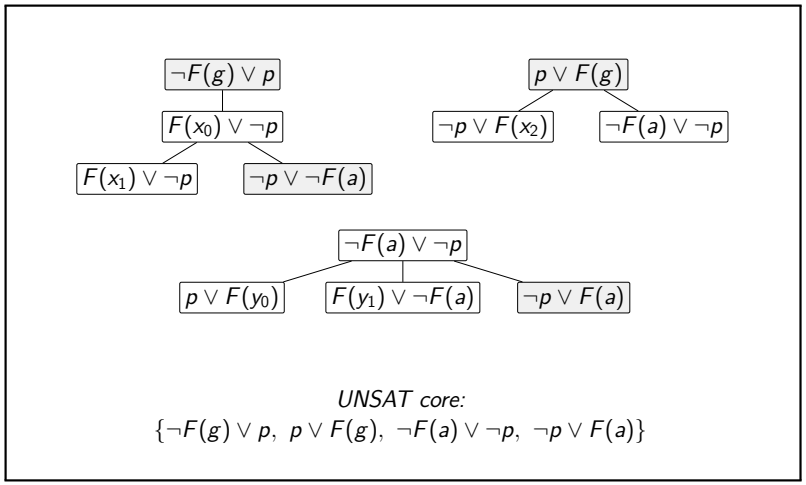
SATCoP: unsat proof in **17 steps** by saturating the SAT core.

Michael Rawson and Giles Reger (2021). "Eliminating Models During Model Elimination". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Vol. 12842. Cham: Springer International Publishing, pp. 250–265

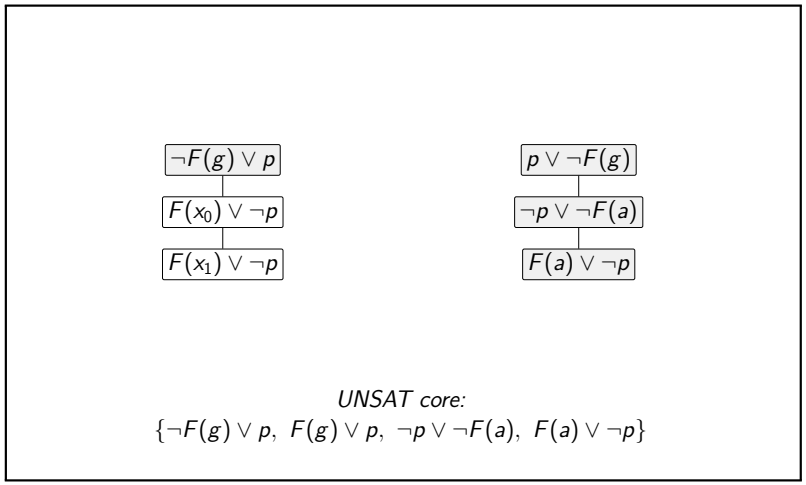
SATResetCoP

Backtracking dominates proof search in connection tableau (Färber 2024)

What if we run Reset every time instead of Backtrack to saturate the UNSAT core faster?



SATCoP: unsat proof in **17 steps** by saturating the SAT core.

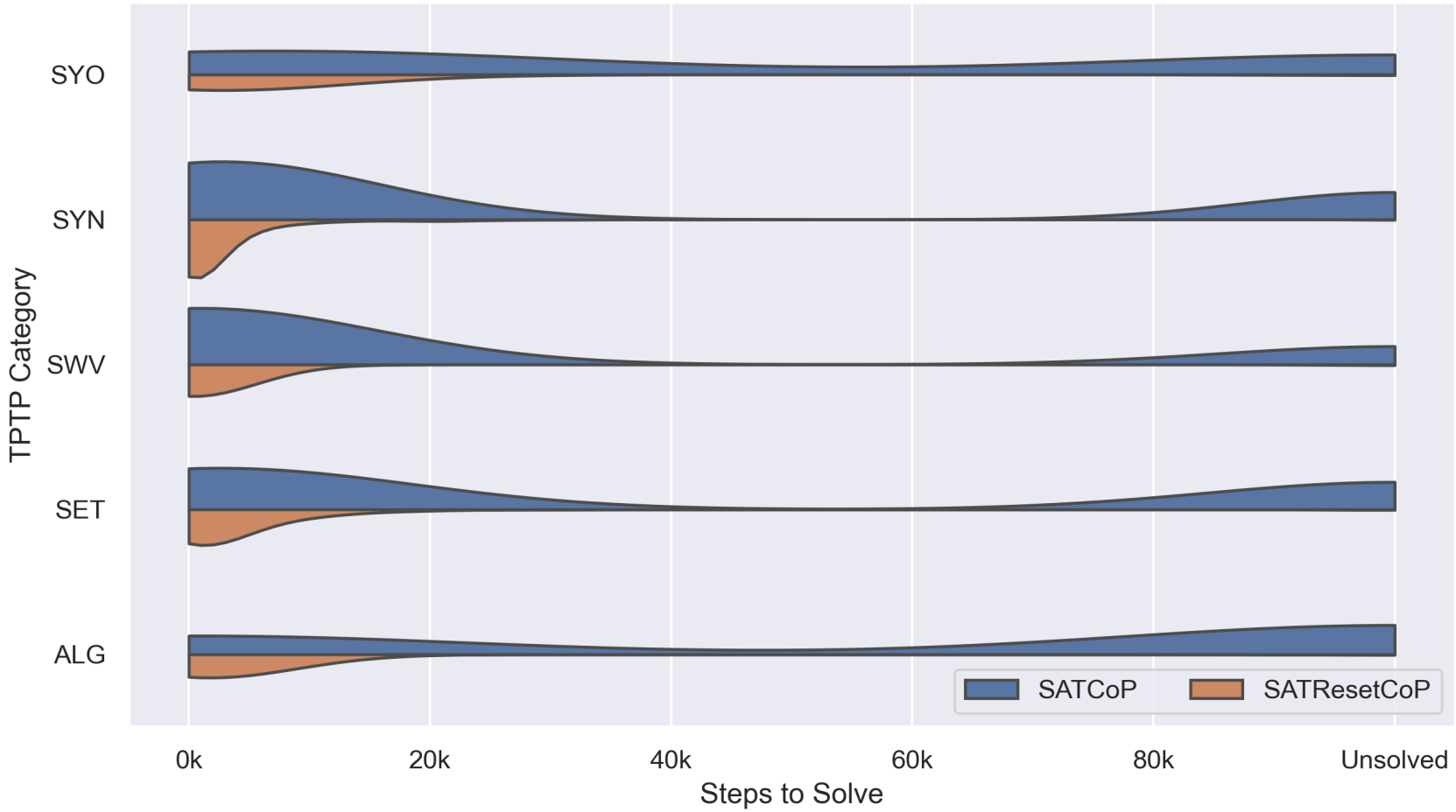


SATResetCoP: unsat proof in **6 steps** by saturating the SAT core.

Michael Färber (Jan. 16, 2024). *A Curiously Effective Backtracking Strategy for Connection Tableaux*. arXiv: 2106.13722 [cs]. URL: http://arxiv.org/abs/2106.13722 (visited on 02/02/2026). Pre-published

SATResetCoP

This turns out to be better in all TPTP categories.



SATResetCoP improves the SATCoP baseline by 24% when testing with a 90 second time limit and 100k step limit.